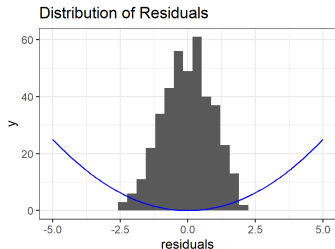
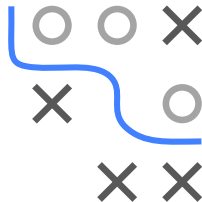


Introduction to Machine Learning

Advanced Risk Minimization Maximum Likelihood vs. ERM



Learning goals

- Max. lik. and ERM are the same
- Gaussian errors = L2 loss
- Laplace errors = L1 loss
- Bernoulli targets vs. log loss

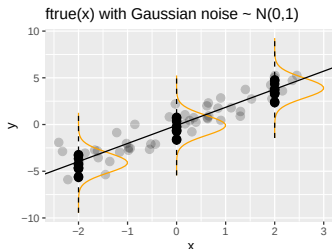
MAXIMUM LIKELIHOOD

- Regression from a maximum likelihood perspective
- Assume data comes from $\mathbb{P}_{xy}$
- Conditional perspective:

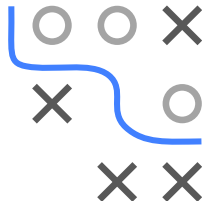
$$y \mid \mathbf{x} \sim p(y \mid \mathbf{x}, \theta)$$

- Common case: true underlying relationship f_{true} with additive noise (surface plus noise model):

$$y = f_{\text{true}}(\mathbf{x}) + \epsilon$$



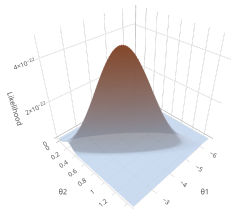
- f_{true} has params θ and $\epsilon \sim \mathbb{P}_\epsilon$, with $\mathbb{E}[\epsilon] = 0, \epsilon \perp\!\!\!\perp \mathbf{x}$
- We now want to learn f_{true} (or its params)



MAXIMUM LIKELIHOOD

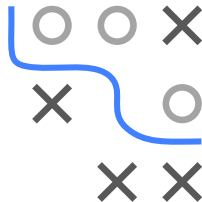
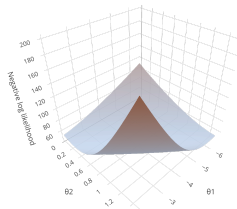
- Given i.i.d data $\mathcal{D} = ((\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)}))$ from $\mathbb{P}_{xy}$
- Max. likelihood maximizes **likelihood** of data under params

$$\mathcal{L}(\theta) = \prod_{i=1}^n p(y^{(i)} | \mathbf{x}^{(i)}, \theta)$$



- Equivalent: minimize **negative log-likelihood (NLL)**

$$-\ell(\theta) = -\sum_{i=1}^n \log p(y^{(i)} | \mathbf{x}^{(i)}, \theta)$$



RISK MINIMIZATION

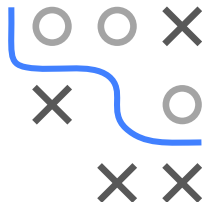
- In ML / ERM: instead of conditional distribution, pick a loss
- Our admissible functions $f(\mathbf{x})$ come from hypothesis space $\mathcal{H}$
- But in stats, must assume some form of f_{true} , no difference
- Simply define neg. log-likelihood as **loss function**

$$L(y, f(\mathbf{x} \mid \theta)) := -\log p(y \mid \mathbf{x}, \theta)$$

- Then, maximum-likelihood = ERM

$$-\ell(\theta) = \mathcal{R}_{\text{emp}}(\theta) = \sum_{i=1}^n L(y^{(i)}, f(\mathbf{x}^{(i)} \mid \theta))$$

- NB: When only interested in minimizer, we use $\propto$ as “proportional up to pos. multiplicative and general additive constants”



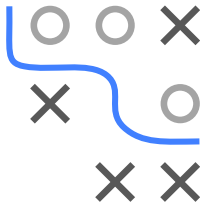
GAUSSIAN ERRORS - L2-LOSS

- Assume $y = f_{\text{true}}(\mathbf{x}) + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$
- Then $y \mid \mathbf{x} \sim \mathcal{N}(f_{\text{true}}(\mathbf{x}), \sigma^2)$ and likelihood is

$$\begin{aligned}\mathcal{L}(\theta) &= \prod_{i=1}^n p(y^{(i)} \mid f(\mathbf{x}^{(i)} \mid \theta), \sigma^2) \\ &\propto \prod_{i=1}^n \exp\left(-\frac{1}{2\sigma^2}(y^{(i)} - f(\mathbf{x}^{(i)} \mid \theta))^2\right)\end{aligned}$$

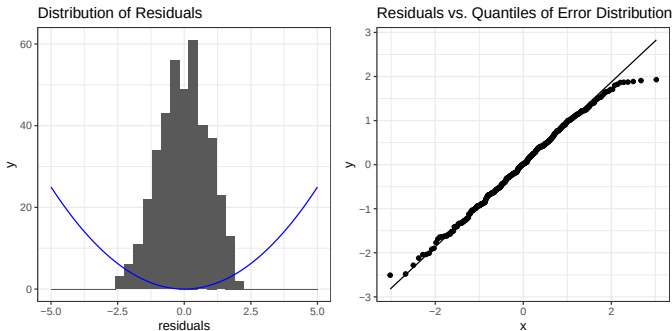
- Minimizing Gaussian NLL is ERM with L2-loss

$$\begin{aligned}-\ell(\theta) &= -\log(\mathcal{L}(\theta)) \\ &\propto -\log\left(\prod_{i=1}^n \exp\left(-\frac{1}{2\sigma^2}(y^{(i)} - f(\mathbf{x}^{(i)} \mid \theta))^2\right)\right) \\ &\propto \sum_{i=1}^n (y^{(i)} - f(\mathbf{x}^{(i)} \mid \theta))^2\end{aligned}$$

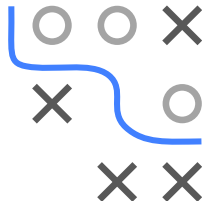


GAUSSIAN ERRORS - L2-LOSS

- Simulate data $y \mid x \sim \mathcal{N}(f_{\text{true}}(x), 1)$ with $f_{\text{true}} = 0.2 \cdot x$
- Plot residuals as histogram, after fitting LM with L_2 -loss (blue)
- Compare emp. residuals vs. theor. quantiles via Q-Q-plot



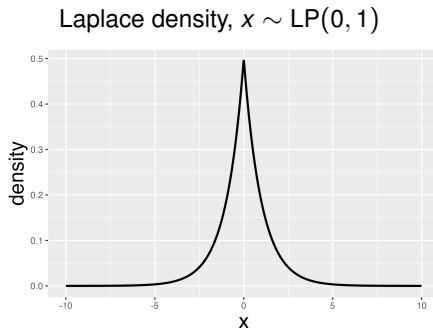
- Residuals are approximately Gaussian!



LAPLACE ERRORS - L1-LOSS

- Consider Laplacian errors ϵ , with density

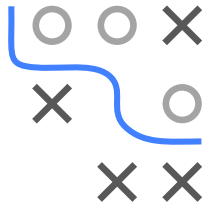
$$\frac{1}{2\sigma} \exp\left(-\frac{|\epsilon|}{\sigma}\right), \sigma > 0$$



- Then

$$y = f_{\text{true}}(\mathbf{x}) + \epsilon$$

also follows Laplace distribution with mean $f(\mathbf{x}^{(i)} | \boldsymbol{\theta})$ and scale σ



LAPLACE ERRORS - L1-LOSS

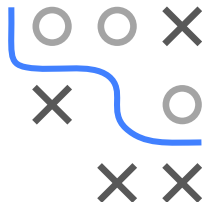
- The likelihood is then

$$\begin{aligned}\mathcal{L}(\theta) &= \prod_{i=1}^n p(y^{(i)} \mid f(\mathbf{x}^{(i)} \mid \theta), \sigma) \\ &\propto \exp\left(-\frac{1}{\sigma} \sum_{i=1}^n \left|y^{(i)} - f(\mathbf{x}^{(i)} \mid \theta)\right|\right)\end{aligned}$$

- The negative log-likelihood is

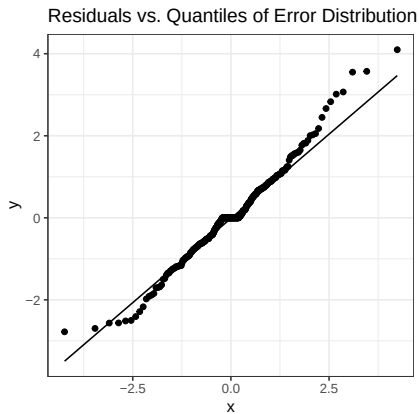
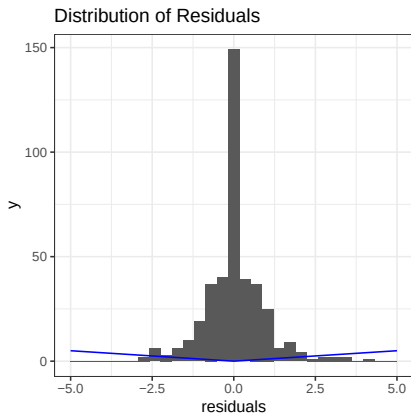
$$-\ell(\theta) \propto \sum_{i=1}^n \left|y^{(i)} - f(\mathbf{x}^{(i)} \mid \theta)\right|$$

- MLE for Laplacian errors = ERM with L1-loss
- Some losses correspond to more complex or less known error densities, like the Huber loss [► Meyer 2021](#)
- Huber density is (unsurprisingly) a hybrid of Gaussian and Laplace

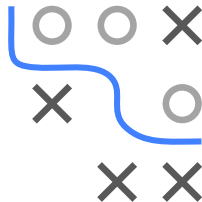


LAPLACE ERRORS - L1-LOSS

- Same setup, now with $y \mid x \sim \text{LP}(f_{\text{true}}(x), 1)$
- Now fit LM with L1 loss



- Again, residuals approximately match quantiles!



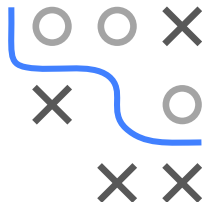
MAXIMUM LIKELIHOOD IN CLASSIFICATION

- Now binary classification
- $y \in \{0, 1\}$ is Bernoulli, $y \mid \mathbf{x} \sim \text{Bern}(\pi_{\text{true}}(\mathbf{x}))$
- NLL:

$$\begin{aligned} -\ell(\boldsymbol{\theta}) &= -\sum_{i=1}^n \log p\left(y^{(i)} \mid \mathbf{x}^{(i)}, \boldsymbol{\theta}\right) \\ &= -\sum_{i=1}^n \log \left[\pi(\mathbf{x}^{(i)})^{y^{(i)}} \cdot (1 - \pi(\mathbf{x}^{(i)}))^{(1-y^{(i)})}\right] \\ &= \sum_{i=1}^n -y^{(i)} \log[\pi(\mathbf{x}^{(i)})] - (1 - y^{(i)}) \log[1 - \pi(\mathbf{x}^{(i)})] \end{aligned}$$

- Results in Bernoulli / log loss:

$$L(y, \pi(\mathbf{x})) = -y \log(\pi(\mathbf{x})) - (1 - y) \log(1 - \pi(\mathbf{x}))$$



DISTRIBUTIONS AND LOSSES

- For **every** error distribution $\mathbb{P}_\epsilon$, can derive an equivalent loss
- Leads to same point estimator for θ as maximum-likelihood:

$$\hat{\theta} \in \arg \max_{\theta} \mathcal{L}(\theta) \Leftrightarrow \hat{\theta} \in \arg \min_{\theta} -\log(\mathcal{L}(\theta))$$

- But **cannot** derive a pdf/error distrib. for every loss, e.g., Hinge loss; some prob. interpretation still possible ▶ Sollich 1999
- For dist.-based loss on residual $L(y, f(\mathbf{x})) = L_{\mathbb{P}}(r)$, ERM is fully equiv. to max. conditional log-likelihood $\log(p(r))$ if
 - 1 $\log(p(r))$ is affine trafo of $L_{\mathbb{P}}$ (undoing the $\propto$):
 $\log(p(r)) = a - bL_{\mathbb{P}}(r)$, $a \in \mathbb{R}, b > 0$
 - 2 p is a pdf (non-negative and integrates to one)

